

ASM Transformation Service

End-to-End Process Map

Converting Laboratory Instrument Data to Allotrope Simple Model (ASM) Format

Powered by AWS Bedrock Claude • Allotropy Library • Custom Converters

Service Overview

A complete pipeline for converting raw instrument files to validated, integrity-verified ASM output.

- Hybrid approach: proven allotropy parsers + AI generation for unknown formats
- 31 instruments supported natively via allotropy library
- Custom converter registry for proprietary instrument formats
- AI-powered fallback using AWS Bedrock Claude 3.5 Sonnet
- DVaaS validation with PDF attestation reports
- Data integrity verification: field-by-field source-to-ASM proof
- FDA 21 CFR Part 11 and EMA Annex 11 compliance ready
- No data persistence — files processed in-memory only

Two Entry Points

Entry Point 1: CONVERT

"I have a raw instrument file and need ASM output"

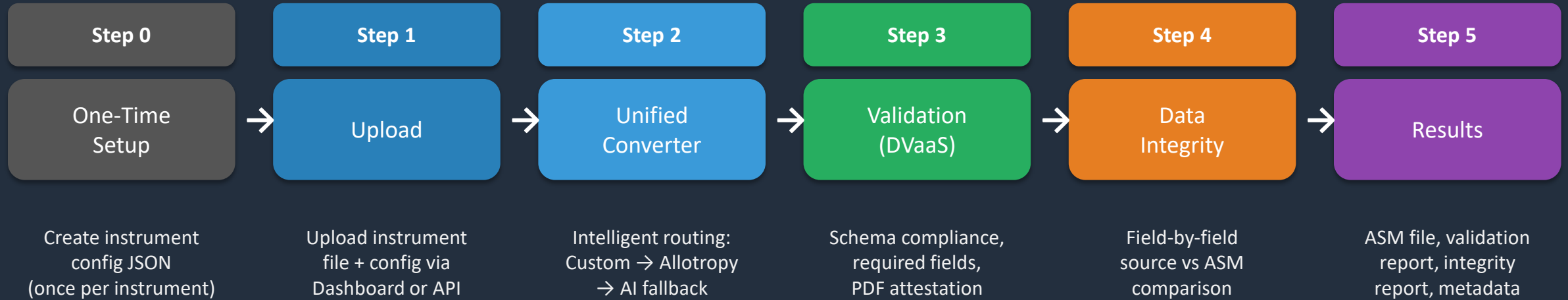
- Provide: Instrument File + Instrument Config (JSON)
- Full pipeline: Upload → Route → Convert → Validate → Integrity Check
- Returns: ASM file + Validation PDF + Integrity report

Entry Point 2: VALIDATE

"I already have an ASM file and want to check compliance"

- Provide: ASM JSON file only
- DVaaS validates any ASM file regardless of source
- Returns: Validation status + errors/warnings + PDF attestation

End-to-End Process Flow



Step 2: Unified Converter — Intelligent Routing

The Unified Converter evaluates routes in priority order and uses the first one that succeeds.

Route A: Custom Converter

- Highest priority
- Exact match: vendor + model
- Registry or embedded converters
- Includes field_mapping ✓
- e.g. Nova FLEX2, EndoScan-V

Route B: Allotropy Library

- Second priority
- 31 supported instruments
- Rule-based parsing
- Field mapping: planned
- e.g. Vi-CELL, SoftMax Pro

Route C: ATaaS (AI)

- Fallback when A & B fail
- AWS Bedrock Claude 3.5 Sonnet
- Analyzes unknown formats
- Generates Python converter
- Best for simple formats

Priority: 1. Custom Registry → 2. Embedded Custom → 3. Allotropy (31 instruments) → 4. ATaaS (AI) → 5. Error

Step 3: Validation — DVaaS

Data Validation as a Service checks ASM structural compliance against Allotrope standards.

Check	Description	Severity
Schema compliance	Does ASM follow Allotrope structure?	Error
Required fields	manifest, measurement docs, sample docs present?	Error
Measurement IDs	UUID v4 format?	Error
Timestamps	ISO 8601 with timezone?	Error
Data source traceability	Calculated values linked to source?	Error
Unit recognition	Standard Allotrope units?	Warning
Equipment metadata	Serial number, software version?	Warning
Device control	Device type, detection type?	Warning

Validation Levels:

Basic (quick structural check) • Comprehensive (production use) • Certification (regulatory submission)

Step 4: Data Integrity Verification

Proves every value from the source instrument file was preserved exactly in the ASM output.
No rounding, no silent data loss, no transformation.

- Converter emits field_mapping array during conversion
- Each entry: source field → source value → ASM field → ASM value → unit
- Dashboard compares source_value to asm_value for every entry
- Result: "100% Data Integrity Confirmed" or list of mismatches

Implementation Status:

Nova FLEX2 (embedded)	✓ All 40 CSV columns
Custom Converter Registry	Spec complete, not yet enforced
Allotropy Library (31 instruments)	Planned
ATaaS (AI)	Not planned

Converter Lifecycle — Adding New Instruments

1. IDENTIFY

Customer has a new instrument
not currently supported

2. CHECK

Is it in allotropy?
(31 instruments)

3. BUILD

Follow Custom Converter
Requirements doc

4. REGISTER

POST /register
Status: PENDING

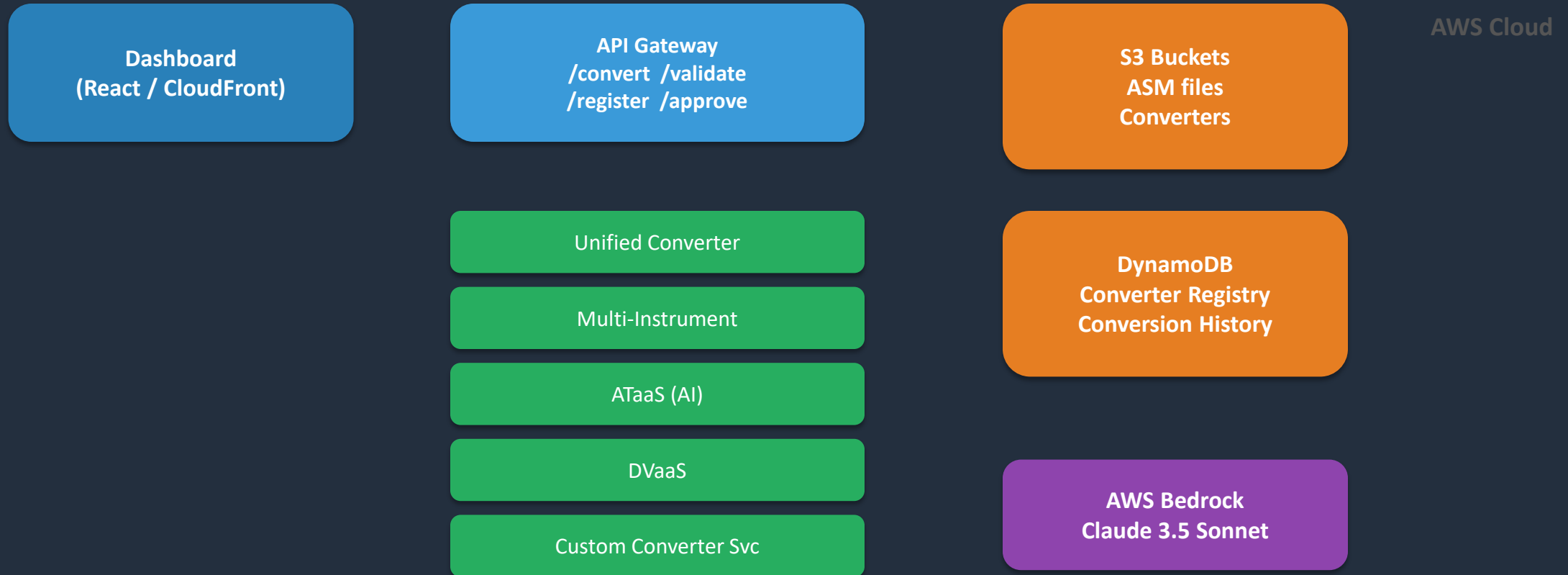
5. APPROVE

Human reviews code
POST /approve

6. AVAILABLE

Auto-used by Unified
Converter for matching files

Service Architecture



Live Endpoints

All services are deployed and operational. Use the Unified Converter for automatic routing.

Service	Endpoint	Purpose
Unified Converter	tqzatn5bse...amazonaws.com/prod/convert	Single entry point (recommended)
DVaaS	4ndgjn16zd...amazonaws.com/prod/validate	ASM validation & certification
Custom Converter	tfv79s08rl...amazonaws.com/prod/	Register, approve, execute
ATaaS	3dbnsq6w6h...amazonaws.com/prod/convert	AI-powered conversion (fallback)
Multi-Instrument	6uogqq4zb5...amazonaws.com/prod/convert	Allotropy library (31 instruments)
Dashboard	d2630v5zyoh8t7.cloudfront.net	Web UI

Dashboard Capabilities

Validate ASM File

Validate an existing ASM file against Allotrope standards

Convert Instrument File

Convert raw instrument file to validated ASM output

Control Tower

Monitor conversion jobs, KPIs, and system health

Instrument Config Creator

Create instrument config JSON files (one-time setup)

Instrument Registry

Browse all supported instruments and their status

Converter Management

Register, review, and approve custom converters

Frequently Asked Questions

Q: What if my instrument is supported by allotropy but I want field_mapping?

A: Register a custom converter. Custom converters take priority over allotropy and include field_mapping.

Q: Do I need a different instrument config for each file?

A: No. One config per instrument. Reuse it for every file from that instrument.

Q: Can I validate an ASM file without converting?

A: Yes. DVaaS validates any ASM file regardless of how it was created.

Q: What's the difference between DVaaS validation and Data Integrity Verification?

A: DVaaS checks structural correctness. Integrity verification checks faithfulness to source data.

Q: How long does conversion take?

A: Custom converters and allotropy: < 1 second. ATaaS (AI): 5-30 seconds.

Q: Is my data stored?

A: Files are processed in-memory only. No data persisted unless explicitly requested.

ASM Transformation Service

Production-Ready • Regulatory Compliant • AI-Powered

- ✓ 31 instruments via allotropy + unlimited via AI
- ✓ Custom converter registry with human approval workflow
- ✓ DVaaS validation with PDF attestation reports
- ✓ Data integrity verification (field-by-field proof)
- ✓ FDA 21 CFR Part 11 and EMA Annex 11 ready
- ✓ No data persistence — in-memory processing only

Powered by AWS Lambda • AWS Bedrock Claude • Allotropy Library • AWS CDK